

Sidharth Sreeram

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University of Waterloo Mechatronics graduate with hands-on experience building software across robotics, perception, and industrial automation. Strong in Python/C++, PyTorch, and system design.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, MATLAB, SQL, Bash

Frameworks & Libraries: PyTorch, NumPy, OpenCV, Boto3, ROS2, Matplotlib, Flask

Concepts: Instruction Tuning, Multimodal Data Curation, Synthetic Data, Data Pipelines, Automated Testing

Development: Linux, AWS, Git, LaTeX, Markdown, JSON, VSCode, Docker, GitHub, Jira, Agile

EDUCATION

AWS Certified Cloud Practitioner | *In Progress*

June 2026 – Present

University of Waterloo | *B.ASc in Mechatronics Engineering*

Sept 2021 – June 2026

– **Relevant Courses:** Cooperative and Adaptive Algorithms, Intro to Machine Learning

EXPERIENCE

Research Assistant - UW WISE Lab | *Python, PyTorch*

Jan 2026 – Apr 2026

- Researched and developed **Python** logic to organize visual and text data into structured VLM prompt format.
- Aggregated NuScenes-QA, NuCaptions, and NuX datasets with multi-sensor metadata by sample token.
- Reconstructed the **OpenDriveVLA** Stage-2 prompt spec, fusing camera, LiDAR, and CAN-bus ego state.
- Generated **728K** JSONL instruction-tuning samples for camera-vs-LiDAR VLA reliability research.

Software Engineering Analyst - Tamaki Controls | *Python, Java, SQL*

May 2025 – Aug 2025

- Designed an **API** response schema improving cross-service communication between **Java** and **Python**.
- Built 6 customer-facing **REST** endpoints importing and exporting UTF-8 encoded CSV/JSON data.
- Wrote **SQL** queries aggregating source data from the database backing each export endpoint.
- Updated **API documentation** with input/output descriptions and example Python client scripts.

Platform Software Engineer - Ford Motor Company | *Python, C++, GCov*

Sept 2024 – Dec 2024

- Developed a **Python CLI** for the AUTOSAR State Management API in a containerized Linux environment.
- Automated 6-stage state transitions, replacing stage-by-stage manual navigation with one command.
- Drove CLI adoption across **3 engineering teams** for state control in testing and integration.
- Created C++ test suites with GCov, raising line coverage **45% to 94%** and condition coverage **50% to 85%**.

Controls System Engineer - Brock Solutions | *Python, SQL*

Jan 2024 – Apr 2024

- Built SCADA HMI screens visualizing live coil crane, acid bath, and pump operations at a steel facility.
- Wrote Python scripts to simulate tag data across a **27-PLC** system to demonstrate HMI designs to customers

PROJECTS

Cloud-Native ADAS Perception Stack | *Python, AWS, CARLA, Docker, boto3, KITTI*

June 2026 – Present

- **Purpose:** Deployment study testing whether cloud-served detections stay viable in near real-time scenarios.
- **Completed:** Generated a 50K+ frame **KITTI** dataset in **CARLA** (RGB, depth, instance-seg, LiDAR, ego pose): 12 highway scenes, 3 maps, day/night, mixed traffic density; test split held out on an unseen map.
- Held **GPU** spend to single-digit dollars per data collection cycle by rendering headless on **Amazon EC2** g4dn.xlarge spot instance and running CPU-intensive tasks on-premise.
- Built a bounded-memory **Amazon S3** streaming loader (thread-pool prefetch, decode-in-stream) so the full dataset processes on a laptop with flat memory and zero disk staging.
- **Planned:** Detector training and benchmarking (mAP, MOTA/IDF1), cloud training with MLflow, ONNX endpoint benchmarked vs. Raspberry Pi edge, and closed loop demo
- **AI Assisted Workflow:** Developed spec-first with Claude Code as an AI pair programmer, iterating each phase's design doc and implementation through rounds of critique and revision before committing code.